

SMART SEA BELT

Smart Wearable Safety System for Fishermen

Team: CEFABEX Innovations

Rohini College of Engineering and Technology,
Kanyakumari

PROBLEM STATEMENT

- Lack of real-time location tracking during emergencies.
- Delayed rescue response after man-overboard incidents.
- Limited access to affordable wearable safety devices.
- No integrated system for emergency alerts and health monitoring.
- About 100000+ people are losing their life by drowning and getting lost in India.

PROBLEM OBJECTIVE

SOS

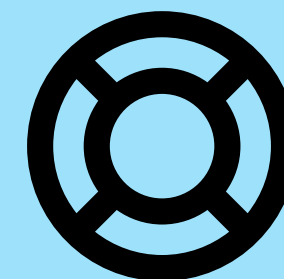
Enable one-touch SOS and automatic emergency notifications.



Provide real-time GPS tracking and IMBL boundary alerts.



Develop a smart wearable safety device for fishermen.



Improve survival and rescue response time during accidents

PROBLEM VALIDATION

- Fishermen face life-threatening risks at sea with delayed help and no real-time alerts.
- Victims may be unable to call for help during emergencies.
- Harsh sea conditions increase the risk of falls, injuries and medical emergencies.
- Real-time location sharing is unavailable in many critical situations.
- Families suffer emotional trauma, financial loss and uncertainty.

REAL WORLD INCIDENT



**300+ fisherman
aressted**



**50+
Boats seized**



**Yearly Average
60+ Incidents**



**Incidents Reported
Across TamilNadu
Indian coastel areas**



**100000+drown every day
during hiking swimming.**

REGISTRATIONS



भारत सरकार
Government of India
सूक्ष्म, लघु एवं मध्यम उद्यम मंत्रालय
Ministry of Micro, Small and Medium Enterprises



सूक्ष्म, लघु एवं मध्यम उद्यम
MSME, SMALL & MEDIUM ENTERPRISES

UDYAM REGISTRATION CERTIFICATE

UDYAM REGISTRATION NUMBER

UDYAM-TN-09-0119792

NAME OF ENTERPRISE

CEFABEX INNOVATIONS

TYPE OF ENTERPRISE *

S.No.	Classification Year	Enterprise Type	Classification Date
1	2026-27	Micro	14/08/2026

MAJOR ACTIVITY

MANUFACTURING

SOCIAL CATEGORY OF ENTREPRENEUR

OBC

NAME OF UNIT(S)

S.No.	Name of Unit(s)
1	CEFABEX INNOVATION

OFFICAL ADDRESS OF ENTERPRISE

Village/Town	RAJAVOOR	Block	AGASTHEESWARAM
Road/Street/Tann	LITTLE FLOWER STREET	City	KANYAKUMARI
State	TAMIL NADU	District	KANNIYAKUMARI , Pin 629402
Mobile	7397107707	Email:	celano.s.12e@gmail.com

DATE OF INCORPORATION / REGISTRATION OF ENTERPRISE

13/08/2026

DATE OF COMMENCEMENT OF PRODUCTION/BUSINESS

S.No.	NIC 2 Digit	NIC 4 Digit	NIC 5 Digit	Activity
1	32 - Other manufacturing	3290 - Other manufacturing n.e.c.	32909 - Manufacture of other articles n.e.c.	Manufacturing

NATIONAL INDUSTRY CLASSIFICATION CODE(S)

DATE OF UDYAM REGISTRATION

14/08/2026

Disclaimer: This is computer generated statement, no signature required. Printed from https://udyamregistration.gov.in & Date of printing:- 14/08/2026

For any assistance, you may contact:

1. District Industries Centre:

KANYAKUMARI (TAMIL NADU)

2. MSME-DFO:

CHENNAI (TAMIL NADU)

Visit : www.msme.gov.in ; www.dcmsme.gov.in ; www.udyamregistration.gov.in

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RAS WATER SPORTS
Adventure | Safety | Water Sports
Manakudi, Tamil Nadu - 695527, India
Email: raswatersports@gmail.com | Phone: +91 94432 67123

BACKWATER TEST REPORT

Evaluation of SMART SEA BELT – Marine Safety & Border Diversion System
(Real Water Environment Test)

Test Conducted By	: RAS Water Sports	Product Name	: SMART SEA BELT
Location	: Backwaters, Manakudi, Kerala	Model / Version	: Prototype v1.0
Test Date	: 28 August 2026	Test Type	: Backwater Field Test
Test Duration	: 3 Hours (09:00 AM – 12:00 PM)	Weather Conditions	: Sunny, Calm Water, 28°C

1. OBJECTIVE

To evaluate the performance, reliability and real-time functionality of the SMART SEA BELT in actual backwater conditions, simulating the environment faced by fishermen during routine operations.

2. TEST SETUP

- 5 Smart Watches (worn by fishermen/volunteers)
- 1 Boat Hub (installed on the boat)
- 1 Base Unit (at shore monitoring station)
- Communication: LoRa (boat hub ↔ base unit)
- Tracking: GPS + NavIC
- Health Monitoring: Heart Rate, Fall Detection
- Emergency: SOS Button

3. TEST RESULTS

S. No.	Parameter Tested	Expected Output	Observed Result	Status
1	GPS / NavIC Location Tracking	Accurate real-time location	Received accurate coordinates (within 5 m accuracy)	Pass
2	Heart Rate Monitoring	Continuous heart rate data	Heart rate displayed (72-96 bpm)	Pass
3	Fall Detection	Alert on sudden fall	Alert triggered within 8 seconds	Pass
4	IMBL Border Detection	Border alert when crossing	Simulated alert received (test zone)	Pass
5	SOS Button	Immediate distress signal	SOS sent to base unit with location	Pass
6	LoRa Communication (Boat Hub ↔ Base Unit)	Stable signal up to 40+ km (in open area)	Stable connection at 3.8 km (in backwater area)	Pass
7	Waterproof Performance	No device failure in water	Devices functioned without leakage	Pass
8	Battery Endurance	Minimum 8 hours	Operated for 8+ hours (90% battery remaining)	Pass

4. CONCLUSION

The SMART SEA BELT performed successfully in backwater conditions. All critical functions including GPS tracking, health monitoring, fall detection, IMBL border alert, SOS and communication worked as expected. The device showed stable performance, good water resistance and reliable battery life, making it suitable for real-world deployment for fishermen operating in backwaters and coastal regions.

Test Conducted By



Team RAS Water Sports
Manakudi, Kerala



Authorized Signatory



Mr. R. Aravind
Director - RAS Water Sports
Manakudi, Kerala

Welcome ROHINICET [Sign out](#)



सत्यमेव जयते
G.A.R.6
[See Rule 22(1)]
RECEIPT



INTELLECTUAL
PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

Docket No 126255

Date/Time 2026/08/17 14:38:00

To

ROHINICET

Userid: ROHINICET

Rohini College of Engineering and Technology Near Anjugramam JN, Kanyakumari Main Road, Palkulam

CBR Detail:

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	202641099158	TEMP-1/10943/2026-CHE	1600	66845	FORM 1	SMART SEA BELT - AN INTEGRATED SMART MARINE SAFETY SYSTEM WITH AUTOMATIC RUDDER-BASED DIVERSION AND

TransactionID	Payment Mode	Challan Identification Number	Amount Paid	Head of A/C No
N-0002031157	Online Bank Transfer	1708260036507	1600.00	1475001020000001

Total Amount : ₹ 1600.00
Amount in Words: Rupees One Thousand Six Hundred Only
Received from ROHINICET the sum of ₹ 1600.00 on account of Payment of fee for above mentioned Application/Forms.
* This is a computer generated receipt, hence no signature required.

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MSME Udyam Registered
UDYAM REG. NO.UDYAM-TN-09-0119792 ENTERPRISE
TYPE Micro — Manufacturing

BACKWATEER TEST - MANAKUDI

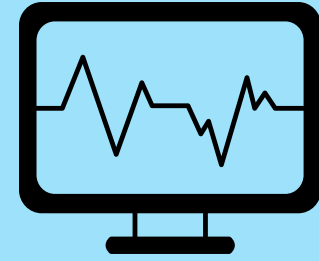
PATENT APPLICATION FILED
APPLICATION NO.202641099158
STATUS FILED — FORM 1 (2026)
filed patent application for an integrated smart marine safety
system with automatic rudder-based diversion

SOLUTIONS



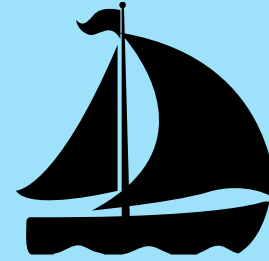
Individual Fisherman Identification

Unique watch IDs (W1, W2, W3...) allow the system to identify and monitor each fisherman separately.



Real-Time Safety Monitoring

The Boat Hub continuously monitors all connected fishermen and detects abnormal or emergency conditions.



Man-Overboard Detection

The system can identify when a fisherman moves outside the predefined safe range of the boat.



Emergency Alert System

Emergency signals from individual watches can be immediately transmitted to the Boat Hub.



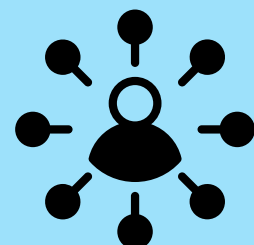
Long-Range Emergency Communication

LoRa enables emergency information to reach the base unit over long distances..



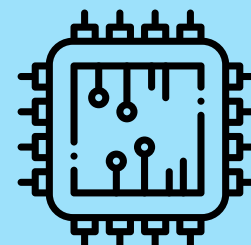
Boat Location Tracking

The Boat Hub provides the vessel's geographical position during normal and emergency situations.



Centralized Monitoring

Multiple fishermen can be monitored simultaneously through a single Boat Hub.



Remote Monitoring Through Cloud

IoT/Cloud connectivity allows shore-based personnel to monitor the fishermen and vessel remotely.



Fisherman-to-Boat Connectivity

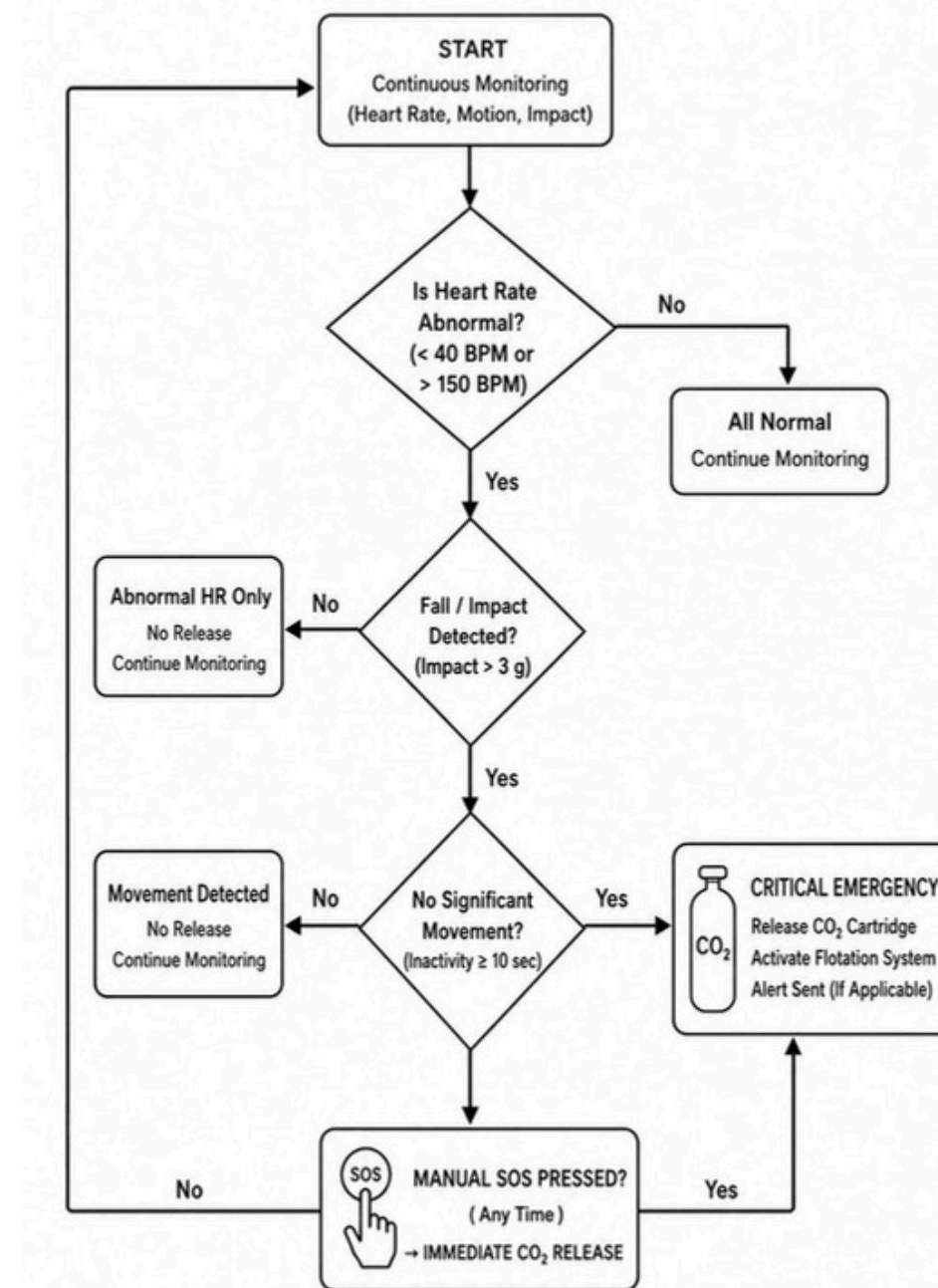
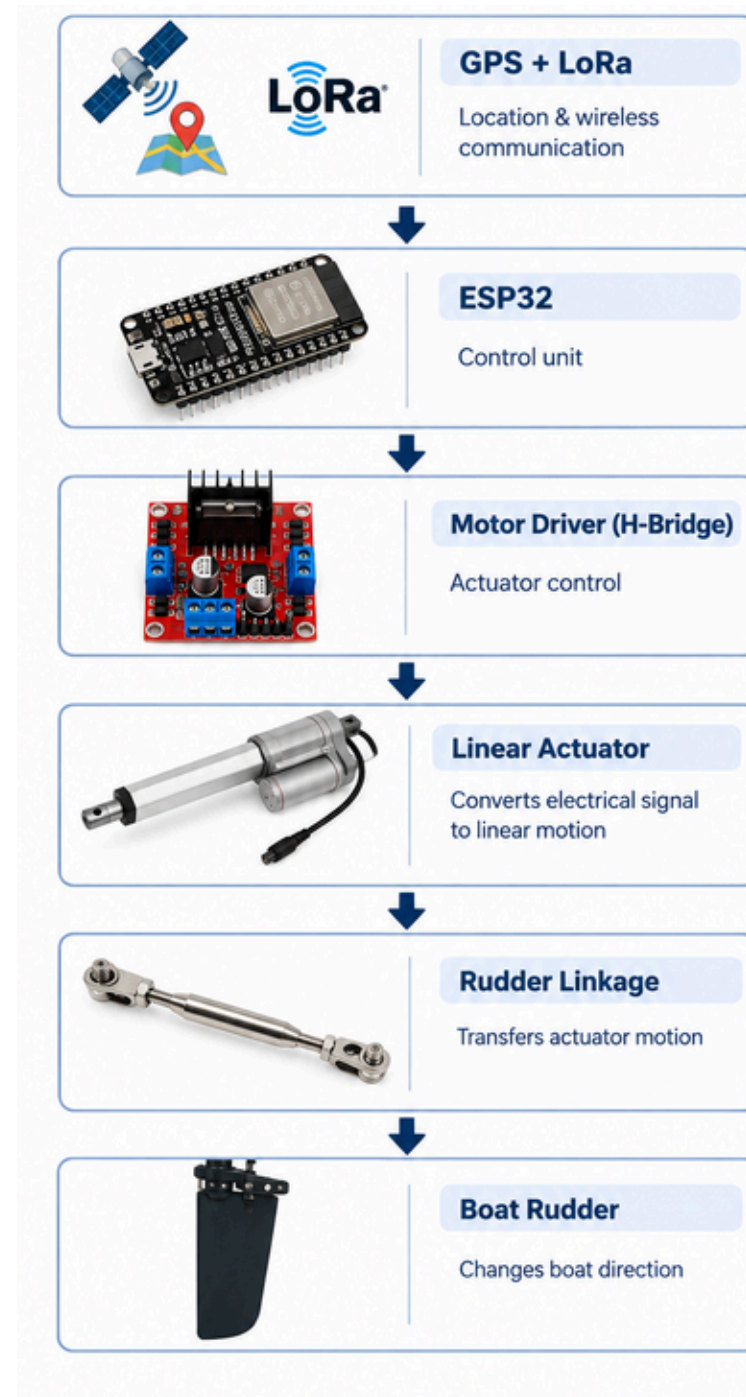
BLE provides low-power communication between individual watches and the Boat Hub.

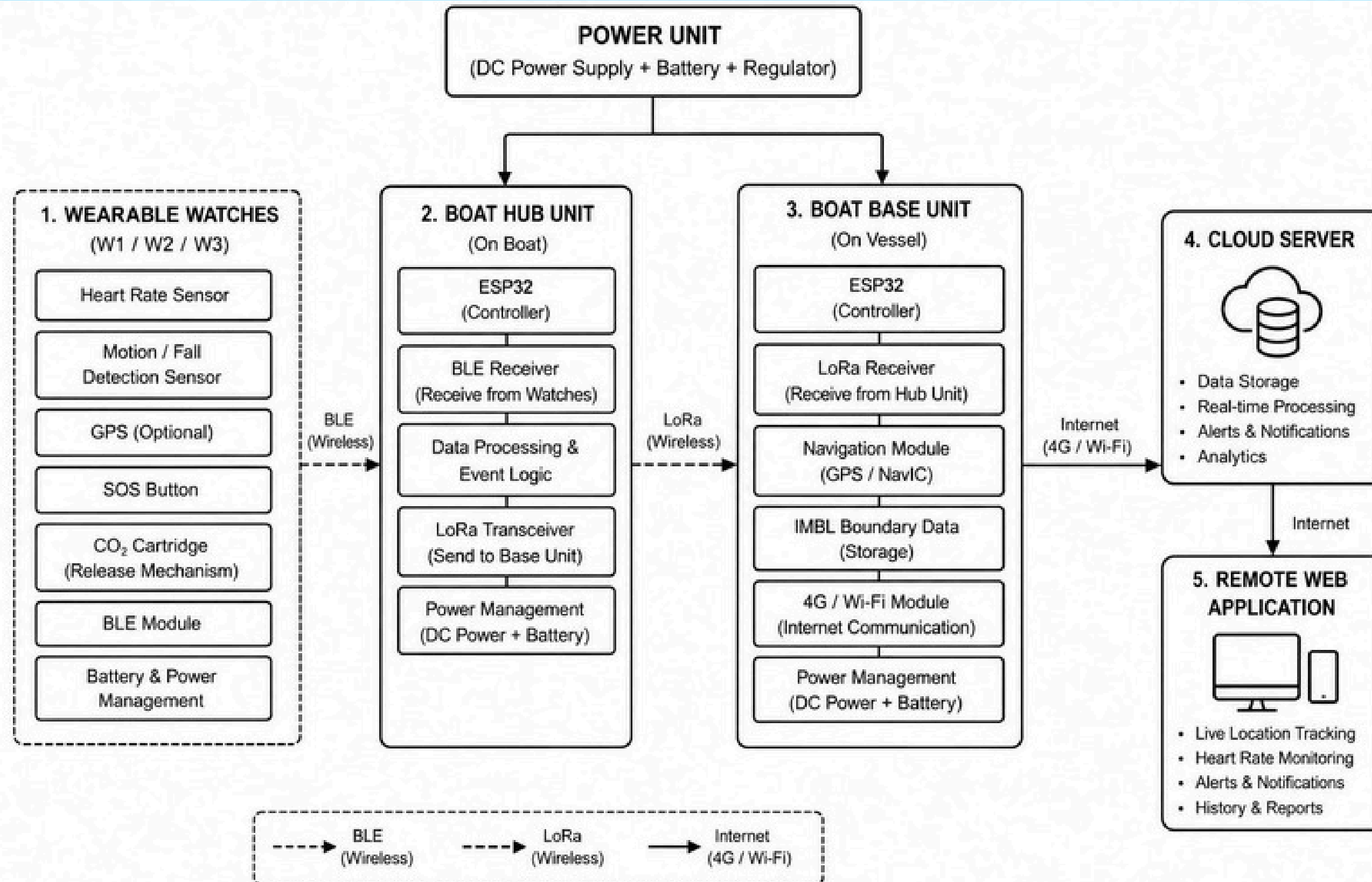


Improved Emergency Response

Combining individual identification, location tracking, emergency alerts, LoRa, and cloud monitoring enables faster response during accidents or distress situations.

TECHNOLOGY ARCHITECTURE







ESP32-C3 + BLE

Central controller with wireless communication



MAX30102 Sensor

Real-time heart-rate monitoring



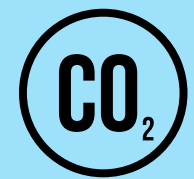
MPU6050 Sensor

Detects falls and abnormal movements



SOS Button

Enables instant emergency alerts



Servo-Based CO₂ Valve

Automatically activates flotation deployment



CO₂ Cartridge + Flotation Device

Provides rapid buoyancy during emergencies



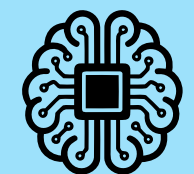
Rechargeable Battery

Powers the wearable system for continuous operation



Automatic Rudder Diversion System

Automatically redirects the boat toward the Indian Ocean.



AI-Integrated Mobile Application

Provides real-time monitoring, emergency alerts, fisherman tracking, and AI-assisted safety analysis.

PROTOTYPE / M V P

TECHNOLOGICAL FEASIBILITY



UNIQUE VALUE PROPOSITION OF OUR PRODUCT



**ALL IN ONE SAFETY
SOLUTION**



**HANDS FREE AUTOMATION
ALERTS**



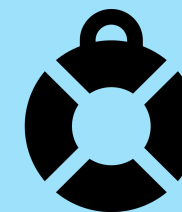
**COMFORTABLE LIGHTWEIGHT
WEARABLE**



**AFFORDABLE & AVAILABLE FOR
EVERYONE**



**AVAILABLE FOR VAST
CUSTOMERS (FISHERMAN,
TRAVELLERS, KIDS, STUDENTS)**



**INCREASES TIMELY
RESCUE & SAVES LIVES**

MARKET OPPORTUNITY

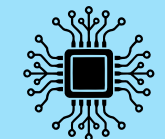
MARKET OPPURTUNITY



RIISING SAFETY CONCERS



**FISHERMEN & COASTAL
COMMUNITIES**



**GROWING ADOPTION
OF WEARABLE DEVICES**



EXPANSION MARKETS

MARKET SIZE

TAM

Total Adressable
Market

N/A

SAM

Serviceable
Available Market

N/A

SOM

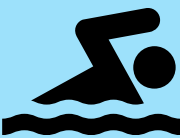
Serviceable
Obtainable Market

N/A

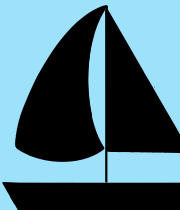
CHANNELS



TOURIST AMBASSADORS



OUTLETS NEAR POOLS



**FISHERMAN & COASTAL
ORGANISATIONS**

INITIAL LAUNCH

TAMIL NADU

KERALA

KARNATAKA

**ANDHRA
PRADHESH**

MARKETING STRATEGY



RENT TO OWN LOCAL AMBASSADORS
**EXPANSION INTO INTERNATIONAL
MARKET**



BUSINESS MODEL & REVENUE STREAM

BUSINESS MODEL



Direct Sales (B2C)



Institutional Sales (B2B)

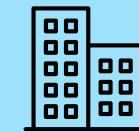


Partnerships

REVENUE STREAM



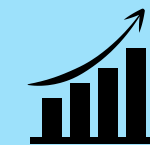
Sales



Bulk orders from companies and institutions.



Licensing the technology to wearable device manufacturers.



Data analytics and safety solutions for enterprise clients



**Collaboration
in Future**

**Kumaran Fishnets Pvt
Nagercoil**

TN Fishnet Initiative

**Porunai Eco Crafters
Thirunelveli**

COST & BUSINESS VIABILITY

BUSINESS VIABILITY



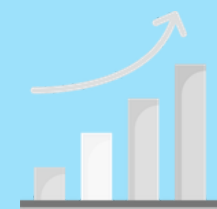
High demand



Multiple Revenue Streams



B2B & B2C



Scalable and profitable

COST STRUCTURE

Component	Estimated Cost (₹)
Base Unit	16,000
Boat Hub	17,000
5 Smart Watches	20,000
Integration & Testing	3,000
Marine Protection / Waterproofing	2,000
Total Prototype Cost	₹58,000

COMPETITIVE ADVANTAGE



Multi-Function Safety Device



Waterproof & Durable



Real-Time Emergency Response



Comfortable Wearable Design



Advanced Diversion System



Easy to Use



Low-Cost & Affordable



Scalable Solution

SOCIAL IMPACT & FUTURE SCOPE

SOCIAL IMPACT



Enhances Fishermen's Safety



Supports Faster Rescue Operations



Improves Family Security



Promotes Safer Fishing Practices



Affordable for Coastal Communities



Reduces Economic Losses

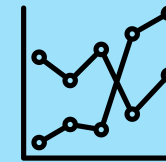


Supports Government & NGO Initiatives

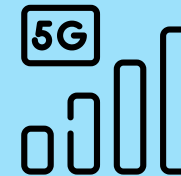


Builds Safer Coastal Communities

FUTURE SCOPE



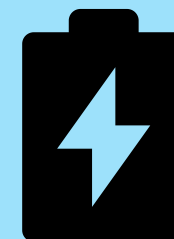
website into application



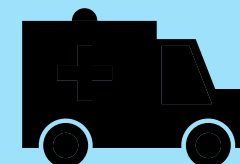
5G/IoT connectivity for faster and more reliable communication.



Multi-language mobile app for wider accessibility.



Longer battery life with energy-efficient hardware and fast charging.



Direct integration with emergency services (112/108) for faster response.

THANK YOU

Insurance covers the loss but we prevent them